

Sergio Fabian Acuña Madrid

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Software developer with hands-on experience building modern web applications using the JavaScript ecosystem — React.js, Next.js, Node.js, and TypeScript. Currently working as a QA Intern, where I have developed End-to-End automation workflows using Playwright that are actively running within a production SaaS platform.

Technical Skills

- **Frontend:** HTML5, CSS3, JavaScript, TypeScript, React, Next.js, Tailwind CSS, Sass/SCSS
- **Backend:** Node.js, Express.js, Python, Java, Spring Boot, PostgreSQL, MariaDB, MySQL, Prisma ORM, RabbitMQ
- **QA & Testing:** Playwright (E2E), Zod, Postman
- **Cloud & Workflow:** Cloudfare, Docker, Vercel, Render, Git/GitHub

Professional Experience

Bitam de México (QA Intern)

Jan 2026 – June 2026

- Developed End-to-End automation workflows using Playwright and TypeScript to replace repetitive manual QA processes, significantly reducing execution time and improving testing efficiency across critical SaaS operations.
- Monitored asynchronous task processing through RabbitMQ queues to validate request delivery, processing order, and execution timing within the production SaaS environment
- Wrote SQL scripts in MariaDB to validate data persistence, verify user profile updates, perform controlled test data cleanup, and ensure backend cURL requests were correctly processed by RabbitMQ queues

Selected Projects

US Accident Risk Predictor (ML-Powered Traffic Analysis Platform)

Nov 2025 – Present

- Engineered a Machine Learning pipeline using Python, Pandas, and Scikit-Learn to process and analyze over 7.7 million historical traffic records, achieving 83.61% predictive accuracy through feature engineering and data preprocessing
- Architected a FastAPI microservice to load and serve a serialized ML model in memory, exposing RESTful endpoints (/predict, /data) for real-time traffic risk inference.
- Designed a multi-service architecture integrating a Python ML engine with a Java + Spring Boot Backend-for-Frontend (BFF), delivering geospatial traffic risk insights to an interactive Leaflet-based map interface.

Bug Tracker MVP (Full-Stack Web App)

Jan 2026 – Present

- Built a role-based ticket management system supporting Admin, Tester, and Developer workflows, with ticket assignment logic and historical change tracking for monitoring issue lifecycles
- Developed analytics dashboards displaying bug trends, unstable modules, reopen rates, ticket distribution by status, critical issue counts, and MTTR metrics to support engineering decision-making
- Implemented CSV export functionality for ticket records, enabling external reporting, data analysis, and workflow auditing

Yama Japanese Food MVP (Mobile-First Web Application)

Jan 2026 – Feb 2026

- Developed a Mobile-First MVP web application for a sushi business, allowing users to filter menu items by preference, view business hours, access store location information, and interact with animated promotional badges
- Achieved 100/100 Google Lighthouse scores across Performance, Accessibility, Best Practices, and SEO on desktop, and 93/100 Performance on mobile through image optimization and responsive design practices.

AI Legislative Assistant (Mobile-First AI Chatbot)

July 2025 – August 2025

- Developed a Mobile-First AI chatbot application for a university student, using prompt engineering techniques to reduce hallucinations and improve response relevance for domain-specific conversations
- Integrated the Google Gemini API into a React.js and Sass/SCSS frontend, while building backend request handling and API communication using Node.js and JavaScript

Education

Instituto Tecnológico de Ciudad Madero

Bachelor's Degree in Software Engineering (Specialization in Big Data)

Cd. Madero, Tamps

Expected August 2026

Languages

Spanish (Native) & English (Advanced).